In [ ]:

```
ML -->
1)SML() -->LR,LgR,KNN,DT,RF,AD,GB
2)UML
3)RML

Model -->?


Pune is the best city.
```

Ridge & Lasso Regression

In [ ]:

```
We have

Area
Bedroom
Bathroom
Location
Age
Distance
Parking
\...


Age   Price

10     100
20     200
30     300
37     400

Overfitting
LargeCoefficients

we Use regularization

it is a technique used to reduce
overfitting by adding a penalty for large model coefficients


Cost=MSE

we minimize:
Cost=MSE + Penalty

2 types:

1)Ridge Regression --->L2 Regularization
2)Lasso Regression --->L1 Regularization
```

Ridge Regression

In [ ]:

```
it is a linear regression technique that add the squared values of coefficients as
a penalty to the cost function

Ridge tries to make coefficient smaller but
generally dose not make them exactly zero

y=m1x1+m2x2+m3x3+mnXn

Age      =12.5
Income   =25.8
Exp      =18.2
Distance =9.5

Ridge

Age --->7.2
Income   =15.4
Exp      =10.5
Distance =4.9


What is Lambda?

Controls the strength of regularization

lambda=0
No Regularization


High Lambda =4
Stronger penalty



If Lambda is too higher the model can underfit


Lambda           Regularization   Coefficient            Risk

0                     None          Large               Overfitting

0.1                 Low              slightly smaller     Lower

1                  Low              smaller              under control

10               High             very small               Underfitting possible

100              Very high          Extremly small          Undefit
```

In [10]:

```
data={
    'Age':[22,25,30,35,40,45,50,28,32,38],
    'Salary':[25000,30000,40000,50000,60000,70000,80000,90000,450000,55000],
    'Experience':[1,2,4,7,10,15,20,3,5,8],
    'Purchased':[0,0,0,1,1,1,1,0,1,1]
}
```

In [12]:

```
import pandas as pd
df=pd.DataFrame(data)
df
```

Out[12]:

	Age	Salary	Experience	Purchased
0	22	25000	1	0
1	25	30000	2	0
2	30	40000	4	0
3	35	50000	7	1
4	40	60000	10	1
5	45	70000	15	1
6	50	80000	20	1
7	28	90000	3	0
8	32	450000	5	1
9	38	55000	8	1

In [16]:

```
Y=df[['Salary']]
X=df.drop(labels=['Salary'],axis=1)
```

In [18]:

```
Y
```

Out[18]:

	Salary
0	25000
1	30000
2	40000
3	50000
4	60000
5	70000
6	80000
7	90000
8	450000
9	55000

In [20]:

```
X
```

Out[20]:

	Age	Experience	Purchased
0	22	1	0
1	25	2	0
2	30	4	0
3	35	7	1
4	40	10	1
5	45	15	1
6	50	20	1
7	28	3	0
8	32	5	1
9	38	8	1

Train Test split

In [23]:

```
from sklearn.model_selection import train_test_split
xtrain,xtest,ytrain,ytest=train_test_split(X,Y,test_size=0.2,random_state=42)
```

Ridge Regression

In [26]:

```
from sklearn.linear_model import Ridge,LinearRegression
from sklearn.metrics import mean_squared_error,r2_score
ridge=Ridge(alpha=1.0)

model=ridge.fit(xtrain,ytrain)

y_pred=model.predict(xtest)

print("MSE: ",mean_squared_error(ytest,y_pred))
print("R2:",r2_score(ytest,y_pred))
```

```
MSE:  80471098600.37509
R2: -0.8247414648611131
```

In [28]:

```
model.coef_
```

Out[28]:

```
array([[ 2281.46008855,  -873.53804665, -9510.94492632]])
```

In [42]:

```
from sklearn.linear_model import Ridge,LinearRegression
from sklearn.metrics import mean_squared_error,r2_score
ridge=Ridge(alpha=10)

model=ridge.fit(xtrain,ytrain)

y_pred=model.predict(xtest)

print("MSE: ",mean_squared_error(ytest,y_pred))
print("R2:",r2_score(ytest,y_pred))
```

```
MSE:  78929715942.85718
R2: -0.7897894771622942
```

In [44]:

```
model.coef_
```

Out[44]:

```
array([[ 1138.39476935,   214.73358997, -1211.19085189]])
```

In [ ]: